APPLIED RESEARCH PROJECT · CAPSTONE PRESENTATION

A Decision-Oriented Conversational Shopping Assistant

Preference Elicitation · Explainable Recommendations · Lifecycle Support

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California State University, East Bay · April 2026

Information Retrieval ≠ Decision Support

Cognitive Overload

Users must mentally juggle all constraints and trade-offs themselves

Vague, Evolving Needs

Users rarely know exactly what they want when they start

Preferences Get Overwritten

Add a new constraint — your earlier choices are erased

Missing Critical Info

Delivery time, promos, and availability hidden or absent

The Decision Burden Falls on the User

Existing tools assume shopping is a search problem.
In reality, users face a decision problem —
and no platform is built to help them solve it.

- Decision fatigue increases with more options
- Users guess, settle, or abandon purchases
- Post-purchase regret when trade-offs aren't clear
- Trust erodes when recommendations have no rationale

→ We built a system that addresses the decision — not just the search.

Existing Platforms — Where They Fall Short

Google Shopping
Search & filter

Amazon Rufus
AI chat in Amazon

ChatGPT Shopping
Natural language AI

Perplexity
AI answer engine

KEY GAPS ACROSS THE LANDSCAPE

- ✗ Overwrites preferences mid-conversation
- ✗ Assumes you already know what you want
- ✗ Missing delivery, promos & availability
- ✗ Text heavy — low information density
- ✗ No preference continuity across turns
- ✗ Multiple steps to reach product page
- ✗ Separates reasoning from shopping flow
- ✗ Opaque ranking — no explanation why
- ✗ Not built for first-time / cold-start users

→ **The common thread: no platform treats shopping as a decision problem.**

Search-first tools
→ retrieve results, not decisions

AI chatbots
→ converse, but don't structure trade-offs

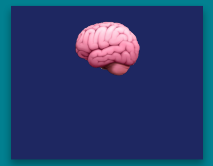
All surveyed platforms
→ no preference continuity across turns

Capability Comparison Across Platforms

Capability	Google Shopping	Amazon Rufus	ChatGPT Shopping	Perplexity	Our System
Preference Elicitation	✗	Partial	Partial	✗	✓
Decision Modeling	✗	✗	✗	✗	✓
Explainable Recommendations	✗	✗	Partial	✓	✓
Delivery / Promo Info	Partial	✗	✗	✗	✓
Preference Continuity	✗	Partial	Partial	✗	✓
Side-by-Side Comparison	✗	✗	✗	✗	✓
Lifecycle Support	✗	✗	✗	✗	✓

→ In our self-evaluation against these platforms, our system addresses all seven gaps. Here is how we built it.

Your Personal AI Shopping Assistant

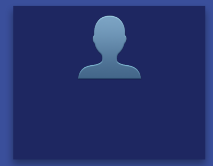


Understand You

Frustration → Category

Decision-Relevant Q&A

ecosystem · placement · privacy · size



Remembers You

Preference Continuity

Visible Diff on Refine

“leaks” → leak-proof · “heavy” → lightweight

Clear Decisions

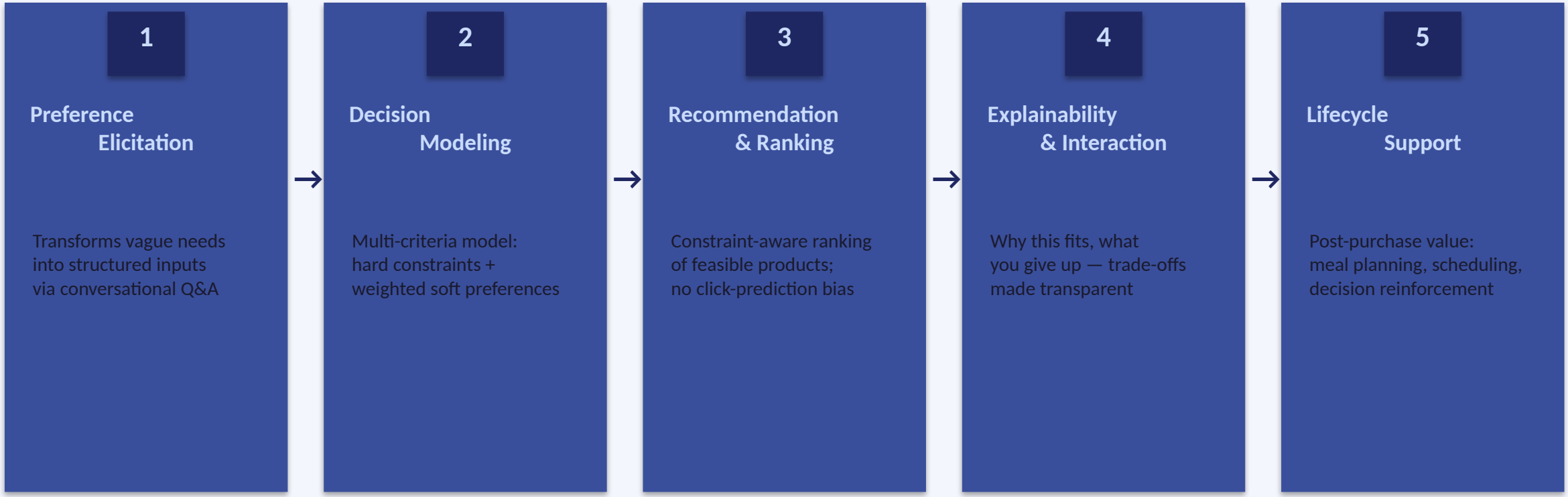
3 Curated Picks

Why-this-fits · Compare

Best Fit · Budget Pick · Stretch Pick

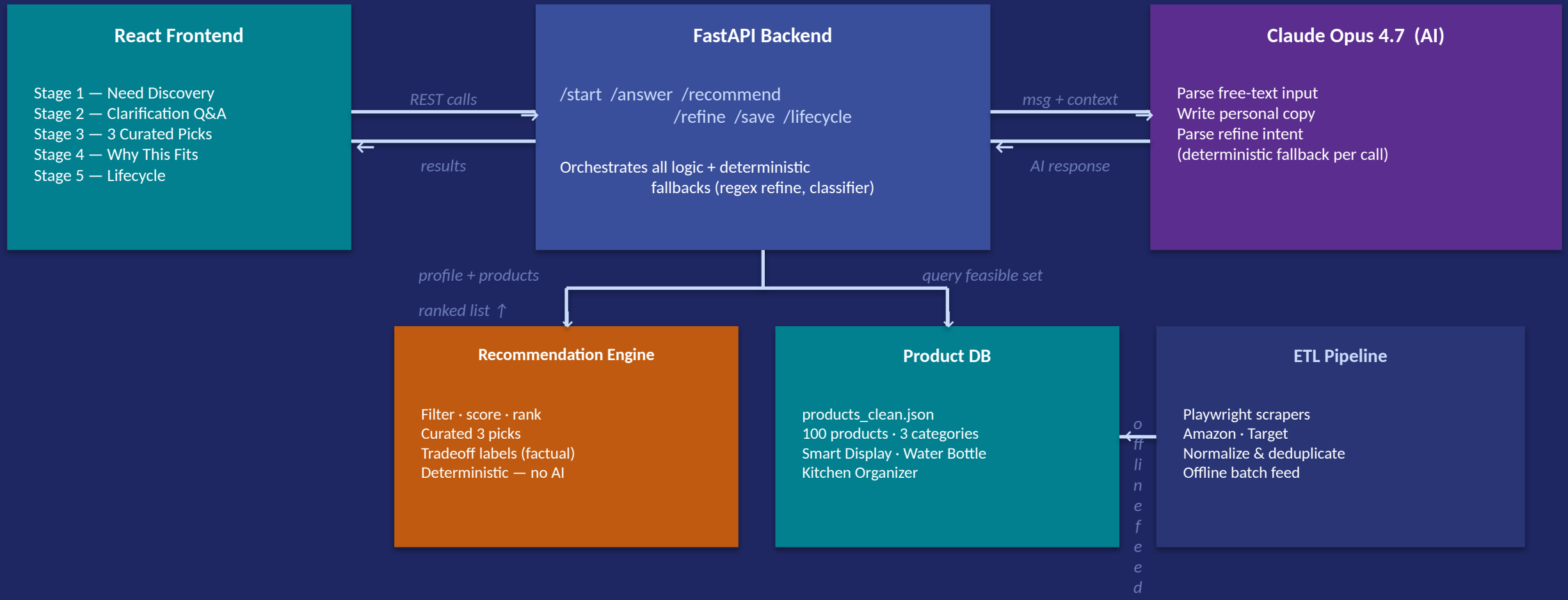
🏠 **Lifecycle Layer (prototype) — Realistic post-purchase support**

Five-Layer Decision Support Framework



End-to-end: vague need → structured decision → ranked products → explained trade-offs → lifecycle value

Components & How They Communicate



The Deterministic Core

Claude — Free-Text Parsing (with regex fallback) | Rule-Based — Ranking · Picks · Labels | Claude — Personal Copy (with grounded fallback)

① Filter

Apply hard constraints (budget, delivery, no-camera).
Non-matches drop out before scoring.

② Score

General: rating · price · delivery (weights from priority)
Feature: category rules (voice · screen · privacy · …)

③ Rank

Combined = General + category bonus.
Sort top-N descending.

④ Curate 3 Picks + Labels

Best Fit · Budget Pick · Stretch.
Factual labels (Cheapest, Most leakproof) — only when true.

Example — User intent: "Reliable smart display · Can wait on delivery · Budget flexible"

GENERAL WEIGHTS — dynamic, derived from this user's priorities

Rating 50%	Price 30%	Delivery 20%

FEATURE WEIGHTS — category-specific (smart displays)

Voice 40%	Screen 30%	Smart Home 20%	Brand 10%

✓ PASS	Amazon Echo Show 8	$79 · 4.7★ · 2-day	Gen: 85 · Feat: 78 → Combined: 82 → #1
✓ PASS	Google Nest Hub (2nd Gen)	$65 · 4.5★ · 3-day	Gen: 78 · Feat: 82 → Combined: 80 → #2
✗ ELIMINATED	Amazon Echo Show 15	$249 · 4.8★ · 2-day	$249 > $80 budget → Eliminated before scoring

When the system must decide what to show the user, AI steps aside. Ranking is rule-based, reproducible, and auditable — trust is anchored here.

The Supporting Stack — What Feeds the Decision Engine

ETL + Provenance Flags

Playwright scrapers · Amazon · Target
Anti-bot stealth · normalize · dedup
Each field flagged: scraped vs estimated
vs synthesised — no silent fabrication

100 products · 3 categories

Attribute Extraction

screen_inches · ecosystems · has_camera
mounting · capacity_oz · bottle_material
drinking_style · organiser_material
Pulled from titles at load time

feeds ranker, picks, compare view

Claude Opus 4.7 — 3 Call Sites

1. Parse free-text input
2. Write personal product copy
3. Parse refine intent

Every call has a deterministic fallback

Never on the ranking path

React Frontend — 5 Stages

1. Need Discovery
2. Clarification Q&A
3. 3 Curated Picks
4. Why This Fits (rule-grounded checklist)
5. Lifecycle (prototype value-extension)

+ Compare overlay · Save-for-later · Diff

AI tooling: Runtime Claude Opus 4.7 · Build Claude Code (Opus 4.7) · Independent validation Codex GPT-5.5

Capability Validation & Scenario Walkthrough

Capability Validation

✓ Validation method (multi-agent)

Claude Code ↔ Codex GPT-5.5 cross-review on the same codebase + family / friends / classmates user testing

✓ Capability gaps (self-evaluation)

7 of 7 gaps addressed in our walkthroughs; no surveyed platform addressed more than 2

✓ Decision-critical attributes inline

Screen size, ecosystem, camera, capacity, material — surfaced on every pick

✓ Trade-offs visible, not hidden

Every pick shows a ✓/✗/? checklist mapping your chips to match status — no surprises

Demo — Water-Bottle Scenario

① Need Discovery

"Current bottle is heavy and leaks in my gym bag." → water_bottle

② Clarification (3 chips)

Gym · Large capacity · (leak inferred from 'leaks' in the input)

③ 3 Curated Picks

Best Fit Owala 24oz FreeSip
rules: gym_suitable · large_capacity_pref · leak_proof_match

④ Picks — labels + checklist

Budget Pick: Zak 19oz [Cheapest · Most portable]
Stretch: Owala 32oz [Most capacity · Most leakproof]

⑤ Refine + Compare

'larger' → flips Best Fit to 32oz · diff card shown
Select 3, compare side-by-side with row-winners

Self-evaluation + small-scale user testing (family · friends · classmates) · formal study is future work (see Slide 11)

Honest Assessment · What This Proves · Next Steps

Limitations

⚠ Small dataset — 100 products, 3 categories only

⚠ Small-scale user testing — family, friends, classmates only

⚠ 72/100 delivery values are estimated (flagged in data)

⚠ Rule-based ranking — weights inferred, not learned

Conclusions & Impact

✓ Decision support feasible end-to-end without an LLM key

✓ Explanations grounded in the same rules the ranker used

✓ Visible preference continuity — every refine shows a diff

✓ Lifecycle layer is a prototype value-extension concept

✓ AI tools disclosed: Claude Opus 4.7 · Codex 5.5 review

Future Work

→ Formal user study (SUS, task completion time)

→ Live retailer API for real-time pricing

→ ML-based ranking from interaction data

→ Expand to more product categories

→ Mobile-first UI redesign

Thank You

Questions & Discussion

Author:	Chenghui Tan
Institution:	California State University, East Bay (CSUEB)

Project:	Decision-Oriented Conversational Shopping Assistant
Runtime:	React · FastAPI · Claude Opus 4.7 · Python · Playwright (ETL)
Built with:	Claude Code (Opus 4.7) · Codex GPT-5.5 (independent validation)